9 Projects That Actually Get You Hired

Real systems-level projects that make recruiters stop scrolling your resume

1. Build Your Own LLM Inference Engine

- Implement tokenization, embeddings, attention, forward pass, inference loop
- Add memory optimization, KV cache, quantization

Stack: C++ / Rust / CUDA

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2. Emotion-Aware Music Recommendation System

- Track mouse movement, clicks, and scroll behavior
- Predict user's emotional state from interaction patterns
- Generate personalized Spotify playlists based on predicted mood

Stack: Python, ML model, Spotify API, frontend event tracking

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3. Real-Time Multiplayer Card Game

- Marvel / Naruto / FIFA themed card battle game
- WebSockets for real-time communication
- State synchronization, matchmaking, concurrency & race condition handling
- Multiplayer backend with microservices

Stack: WebSockets, Node.js/Go backend, Redis, microservices

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4. AI Second-Brain Hardware Device

- ESP32 + LCD + microphone hardware build
- Voice recording → Whisper transcription pipeline
- Local AI using Ollama for processing
- Tasks, reminders, habit / GitHub / gym tracking

Stack: ESP32, Whisper, Ollama, embedded C/C++

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5. Decentralized Peer-to-Peer Messaging App

- Direct device-to-device communication, no central server
- Local data storage with encryption & authentication
- Peer discovery, NAT traversal, connection management

Stack: WebRTC / libp2p, encryption libraries, socket programming

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6. Build Your Own Code Editor / IDE

- Syntax highlighting and real-time error detection (squiggles)
- File explorer, undo/redo
- Piece tables or other advanced text data structures
- Optional: add AI/agentic capabilities

Stack: C++ / Rust / Go

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7. Local-First Git-Based Note-Taking App

- Offline-first architecture with local data storage
- Multi-device sync with Git-like version history
- Advanced text formatting support
- Bonus: try to build a real user base for it

Stack: Local DB (SQLite/CRDT), sync engine, any frontend

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8. Home Lab / Personal Data Center

- Turn an old laptop or Raspberry Pi into a server using Docker + Linux
- Set up a reverse proxy
- Self-hosted Google Drive alternative, media server, VPN, ad blocker, password manager
- Run local AI on the same box

Stack: Docker, Linux, Nginx/Caddy, self-hosted apps (Nextcloud, Jellyfin, Pi-hole)

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9. 3D Path Tracer / Ray Tracing Engine

- Build a 3D renderer from scratch using core math
- Implement shadows, reflections, refraction, diffuse lighting
- Physically based materials and full path tracing
- Multithreading, memory management, BVH/spatial data structures
- Optional: move it to the GPU

Stack: C++, GPU compute (CUDA/OpenCL) for the optional GPU version

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Pick one, build it in public, document the process. Depth on one project beats breadth on ten.